Notes: Titman, Wei, and Zhao (JFE, 2022)

Corporate Actions and the Manipulation of Retail Investors in China: An Analysis of Stock Splits

Contents

1 Big picture

- **Question.** Can corporate insiders use a salient corporate action, specifically stock splits, to temporarily inflate stock prices and transfer losses to less sophisticated retail investors?
- **Setting.** The paper studies Chinese A-share stock split announcements from 1999 to 2015, using event returns, confidential Shanghai Stock Exchange (SSE) account-level trading data, off-exchange block trades, share pledge loans, and Eastmoney Guba message-board data.
- **Core mechanism.** Stock splits attract retail investor attention. In a market dominated by retail trading and constrained short selling, this attention can generate temporary price pressure. Insiders or informed investors can then sell into the inflated demand or use the high price as collateral.
- **Main empirical strategy.** The authors identify an ex ante group of "suspicious" stock splits using publicly observable warning signs: impending lock-up expirations, atypical timing, and high accruals. They then compare returns, trading flows, insider-related transactions, and message-board activity around suspicious versus nonsuspicious splits.
- **Main result.** Suspicious splits display an inverse U-shaped return pattern: prices rise around the split announcement and in the next few months, then reverse strongly over the following 15 months. Regular splits and SOE splits do not show the same reversal.
- **Investor-side result.** Small retail investors are net buyers of suspicious splits, while large accounts and institutional investors sell after the announcement. Account-level tests show that suspicious split buyers tend to have smaller accounts, weaker performance, less experience, high trading frequency, and are more likely to be male.
- **Insider-side result.** Suspicious splits are associated with subsequent block trades and share pledge loans, suggesting that insiders and influential shareholders can benefit from the temporary price inflation.
- **Economic takeaway.** A corporate action that is value-neutral in accounting terms can become a manipulation device when it attracts retail attention and arbitrage is limited.

2 Data and measurement (key objects)

2.1 Stock split sample and return data

- **Main market.** Chinese A-shares listed on the Shanghai and Shenzhen stock exchanges.
- **Main event.** Stock split announcements. In China, firms can issue either stock dividends or stock transfers; both are effectively stock splits and do not directly change firm operations or earnings.
- **Initial sample.** The authors begin with 4,510 stock split announcements with complete accounting information and at least one year of prior returns.
- **Final event sample.** After excluding observations with trading halts or abnormal financial conditions, the main sample contains 3,716 stock split announcements.

- **Sample period.** Split announcements run from January 1999 to June 2015. Postannouncement return analysis extends through December 2016.
- **Return source.** Daily stock returns and split announcement data come from CSMAR.
- **Benchmark.** Main abnormal returns are calculated relative to size-decile value-weighted benchmark portfolios.

2.2 Trading-account data

- The authors use confidential SSE account-level trading records for stock transfer events from January 1999 to December 2015.
- The account data contain encrypted account identifiers, security code, trade price, trade volume, trade direction, and transaction date/time.
- A second full-SSE trading dataset from 2013 to 2015 covers all stocks and is used for robustness and account-characteristics tests.
- Investor groups are constructed from SSE account categories:
  - **small retail**: retail accounts with account wealth below RMB 5 million;
  - **large accounts**: accounts with wealth at least RMB 5 million;
  - **institutional investors**: mutual funds, broker asset-management accounts, broker proprietary accounts, insurance companies, and other institutions;
  - **other investors**: qualified foreign institutional investors, social security accounts, and related categories.
- Retail investors dominate trading: small and large retail accounts together account for roughly 89% of trading volume, with small accounts alone accounting for roughly 60%.

2.3 Off-exchange transactions and message boards

- **Block trades.** Off-exchange block transactions from CSMAR, available from 2002 to 2015. A block trade is a transaction larger than RMB 2 million or 300,000 shares.
- **Share pledge loans.** WIND data from 2006 to 2015 identify loans obtained by pledging company shares as collateral.
- **Lock-up expirations.** WIND share issuance data are used to identify restricted shares becoming tradable around split announcements.
- **Message-board data.** Eastmoney Guba posts from January 2010 to March/April 2013 are used to measure retail discussion before suspicious splits. The final message-board sample has 789,461 posts on 1,410 stocks.

2.4 Definition of suspicious splits

For non-SOE split announcers, the paper defines a stock split as suspicious if it satisfies at least one of three ex ante warning signs:

1. **Lock-up expiration.** Influential shareholders have restricted shares becoming tradable in months -1 to $+6$ around the split announcement.
2. **Atypical timing.** The firm either announces the split after poor recent performance, defined as being in the bottom quintile of past three-month returns, or announces the split outside the usual earnings-announcement window.
3. **High accruals.** The firm is in the top quintile of accruals, where accruals proxy for potentially rosy accounting numbers.

   A compact way to write the classification is:

$$Suspicious_i = 1\{NonSOE_i = 1\} \cdot 1\{Lockup_i = 1 \text{ or } Atypical_i = 1 \text{ or } HighAccrual_i = 1\}.$$

Interpretation. The suspicious classification is deliberately based on public, pre-event information. This matters because the authors want to avoid labeling events as manipulative only after observing negative returns.

2.5 Key descriptive facts

- Roughly 21% of split announcements are classified as suspicious.
- Split announcers tend to be larger, have higher share prices, greater analyst following, higher recent returns, higher ROA, lower leverage, lower book-to-market ratios, and higher accruals than nonsplit firms.
- Suspicious split firms are not wildly different from nonsuspicious split firms on many observables such as price level, analyst coverage, and turnover.
- Suspicious splits are smaller by market capitalization and, by construction, have poorer recent returns or higher accruals depending on the suspicious category.
- The similarity between suspicious and nonsuspicious split firms on many observables is important: it helps explain why unsophisticated investors may not recognize suspicious warning signs.

2.6 Timing and unobservables

- The main event time is the stock split announcement date/month.
- The paper studies both short-window announcement effects and longer-window return reversals.
- The authors cannot observe insiders' intentions directly. Instead, they infer manipulation from a combination of price reversal, retail buying, sophisticated-investor selling, insider-benefit transactions, and message-board activity.
- The authors also cannot prove that the split alone causes manipulation; their interpretation is that splits are one tool in a broader manipulation toolkit.

3 Models and empirical specifications (all equations + interpretation)

3.1 Event-study abnormal returns

The basic abnormal return object is a buy-and-hold abnormal return:

$$BHAR_{i,[a,b]} = \prod_{t=a}^{b}(1 + R_{it}) - \prod_{t=a}^{b}(1 + R_{it}^{bench}),$$

where:

- R_{it} is the return of stock i;
- R_{it}^{bench} is the corresponding size-decile benchmark return;
- $[a, b]$ is an event window around the split announcement.

The authors compare average $BHAR$ across windows:

$$[-3, -1], \quad [0], \quad [+1, +3], \quad [+4, +18], \quad [0, +18].$$

Interpretation.

- Manipulation should create a temporary price increase followed by a reversal.
- Under this view, the key signature is not just a positive announcement return; it is a positive run-up followed by negative long-horizon abnormal returns.
- Nonsuspicious splits may generate positive drift because of standard underreaction or signaling, but they should not exhibit sharp reversal.

3.2 Retail-attention heterogeneity

The paper studies whether suspicious split returns are stronger when retail attention is likely to matter more. The main conditioning variables are:

- small versus large market capitalization;
- postsplit nominal price below versus above RMB 10;
- unlucky versus non-unlucky listing codes.

The idea can be summarized as:

$$E[BHAR_{i,t} \mid Suspicious_i = 1, Attention_i = High] - E[BHAR_{i,t} \mid Suspicious_i = 1, Attention_i = Low].$$

Interpretation.

- Small stocks and low nominal prices are more attractive to retail investors.
- Listing-code superstition provides a behavioral proxy: if retail investors avoid unlucky codes, suspicious splits with unlucky codes should be harder to manipulate.
- The paper finds stronger inverse U-shaped patterns when retail attention is stronger.

3.3 Trading-flow measure

For each investor group j, stock i, and day t, net buying is measured as:

$$NetBuy_{i,t,j} = \frac{RMBBuy_{i,t,j} - RMBSell_{i,t,j}}{AverageDailyVolume_{i,t-1,j}}.$$

Cumulative net buying over an event window $[a, b]$ is:

$$CumNetBuy_{i,[a,b],j} = \sum_{t=a}^{b} NetBuy_{i,t,j}.$$

Interpretation.

- This measure shows which investor groups supply or demand shares around the split announcement.
- A manipulation story predicts that retail investors buy while sophisticated investors and insiders sell.
- Because the authors observe actual account categories, this evidence is much stronger than inferring retail trading from trade size.

3.4 Account-level test: who buys suspicious splits?

The authors estimate Poisson regressions of the number of suspicious split purchases by individual accounts:

$$E[Y_i \mid X_i] = \exp\left(\alpha + \beta_1 Wealth_i + \beta_2 ReturnPerformance_i + \beta_3 Experience_i + \gamma' \Theta_i\right),$$

where:

- Y_i is the number of suspicious split purchases by account i;
- $Wealth_i$ is log average monthly account value;
- $ReturnPerformance_i$ is account performance excluding suspicious split holdings;
- $Experience_i$ is account age in months;
- Θ_i includes age, gender, number of stocks held, and number of purchases.

Interpretation.

- If less sophisticated investors are more likely to buy suspicious splits, then wealth, performance, and experience should be negatively related to suspicious split purchases.
- High purchase frequency and male gender are interpreted as possible signs of overconfidence, following the behavioral finance literature.

3.5 Beneficiary tests: block trades and pledge loans

The paper estimates monthly panel regressions:

$$Y_{it} = \alpha + \beta_1 SuspiciousSplit_{i,(t,t-1)} + \beta_2 Split_{i,(t,t-1)} + \gamma' \Theta_{i,t-1} + \lambda_{ym} + \delta_{industry \times year} + \varepsilon_{it},$$

where Y_{it} is either:

- the value of block trades as a percentage of market capitalization, or
- the value of new share pledge loans as a percentage of market capitalization.

The decomposed specification replaces *SuspiciousSplit* with the three suspicious components:

$$LockupSplit_{it}, \quad AtypicalSplit_{it}, \quad HighAccrualSplit_{it}.$$

Interpretation.

- A positive coefficient on suspicious splits in the block-trade regression suggests insiders or large shareholders sell after suspicious splits.
- A positive coefficient in the pledge-loan regression suggests insiders exploit higher stock prices to borrow against shares.
- The use of year-month fixed effects and industry-year fixed effects helps control aggregate market conditions and sector trends.

3.6 Message-board activity test

The authors estimate a Poisson model of message-board activity:

$$E[Activity_{it} \mid X_{it}] = \exp\left(\alpha + \beta_1 SuspiciousPreperiod_{it} + \eta'\Theta_{i,t-1} + \chi_i\right),$$

where:

- $Activity_{it}$ is a count measure such as number of title characters, number of post characters, or number of posts;
- $SuspiciousPreperiod_{it} = 1$ if a suspicious split will occur in the next 14 days;
- χ_i are firm fixed effects.

Interpretation.

- If message-board activity rises before suspicious split announcements, the split may be part of a broader pump-and-dump effort.
- Firm fixed effects mean identification comes from abnormal activity within the same firm, not cross-sectional differences in firms' usual online attention.

3.7 Matched benchmark robustness

To address confounding events, the authors construct matched-sample abnormal returns:

$$MatchedBHAR_{i,[a,b]} = BHReturn_{i,[a,b]} - \frac{1}{|\mathcal{M}(i)|}\sum_{k\in\mathcal{M}(i)} BHReturn_{k,[a,b]}.$$

Matched firms are nonsplit firms with similar:

- size and suspicious characteristics;
- size and earnings surprise;

- size and dividend-announcement timing.

  Interpretation.

- This checks whether the return reversal is really about suspicious splits rather than earnings news, dividends, or the suspicious characteristics alone.
- The main inverse U-shaped pattern survives these alternative benchmarks.

4 Identification and instruments (what makes the estimates credible)

4.1 What is hard to identify

- Stock splits are endogenous corporate actions. Firms choose when to split, often around earnings announcements and other news.
- Insiders' motives are not directly observed.
- Retail investors may simply misvalue splits rather than being intentionally manipulated.
- Return reversals alone cannot prove manipulation because reversals could reflect omitted risk, overreaction, or confounding events.

4.2 Why the suspicious classification helps

- Suspicious splits are identified using public, pre-event warning signs rather than ex post poor performance.
- The three suspicious categories are tied to plausible manipulation incentives:
  - lock-up expirations create incentives to sell at temporarily high prices;
  - atypical timing flags splits that are hard to justify by the normal trading-range story;
  - high accruals suggest a broader pattern of rosy information disclosure.
- SOEs are excluded from the suspicious group because government-appointed executives face different political and incentive constraints and have less equity-based motivation to manipulate.

4.3 Why account-level data are central

- The paper observes actual buyer and seller categories instead of using trade size as a noisy proxy.
- Small retail investors buy suspicious splits, while large accounts and institutions sell.
- This trading pattern directly matches the proposed transfer mechanism: less sophisticated investors provide liquidity to more sophisticated sellers.

4.4 Why insider-benefit evidence matters

- If suspicious splits merely reflected retail misvaluation, insiders would not necessarily benefit in systematic ways.
- The association with block trades and pledge loans suggests that insiders and influential shareholders have concrete financial channels for exploiting temporary overvaluation.

- The lock-up expiration result is especially important because it links a split announcement, an impending ability to sell, and subsequent block transactions.

4.5 Why robustness tests are informative

- Alternative matched benchmarks show that the inverse U-shaped pattern is not simply driven by earnings surprises, dividends, or suspicious characteristics alone.
- Message-board evidence provides a separate attention channel.
- Heterogeneity by small capitalization, low nominal price, and unlucky listing codes supports the interpretation that retail attention is a key input into manipulation.

4.6 Limitations

- The paper cannot observe manipulation agreements or intent directly.
- Stock splits may be only one part of a broader manipulation package that includes rumors, message-board activity, accounting choices, and insider transactions.
- The evidence is strongest as a cumulative pattern: price reversal plus retail buying plus sophisticated selling plus insider-benefit channels.
- The Chinese institutional setting, especially retail dominance and short-sale constraints, is central; the findings do not mechanically generalize to all markets.

5 Tables and figures

5.1 Figure 1: abnormal returns around split announcements

Read:

- Figure 1 plots buy-and-hold abnormal monthly returns for four groups: U.S. splits, regular non-SOE splits, SOE splits, and suspicious splits.
- Suspicious splits rise sharply after the announcement and then fall back, producing the main inverse U-shaped pattern.
- Regular non-SOE splits and SOE splits do not display the same long-run reversal.
- This figure provides the visual core of the paper: suspicious splits look different from ordinary split announcements.

5.2 Table 1: summary statistics of split announcers

Read:

- Split announcers are generally larger, have higher prices, stronger recent performance, better profitability, lower leverage, lower book-to-market, and higher accruals than nonsplit firms.
- Suspicious split announcers are smaller than nonsuspicious split announcers but are otherwise not obviously separable on many basic observables.

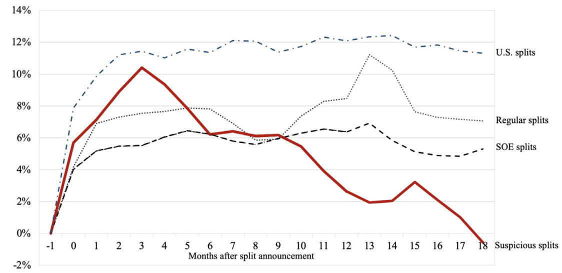


Fig.. 1. Buy-and-hold abnormal monthly returns of split announcements.
Fig. 1 plots the average buy-and-hold abnormal monthly returns for stock splits announced between 1999/01 and 2015/06. The buy-and-hold abnormal return is the buy-and-hold return of the split announcer minus the size-decile return benchmark (DGTW-benchmark for U.S. splits). U.S. splits are all split announcers in the U.S. stock market (dotted-dashed line). Regular splits are split announcers by non-SOEs that are not categorized as suspicious (dotted line). SOE splits are split announcements by SOE firms (dashed line). Suspicious splits are split announcements by non-SOEs that meet one or more of the following criteria: lock-up expirations, atypical timing, or high accruals (solid line).

Table 1
Summary statistics of split announcers.
Table 1 reports the summary statistics of stock splits announcers. Panel A reports the time-series monthly average firm characteristics of firms that announce and did not announce stock splits. Panel B reports the time-series average monthly average firm characteristics of suspicious splits and nonsuspicious split announcers. The last two columns of each panel report the average difference and t-statistic between the two samples. The bottom row reports the time-series annual average number of unique firms in each respective sample. Appendix A provides variable definitions. The sample period is from 1999/01 to 2015/06.

	Split firms	Nonsplit Firms	Difference	t-stat
Panel A. Comparison of split firms and nonsplit firms				
Size (in millions RMB)	$6447	$5298	$1149	(5.85)
Price	$17.57	$12.30	5.27	(12.09)
Analyst	10.2	6.3	3.9	(4.22)
Turnover (qtr)	39.9%	42.6%	−2.7%	(−1.84)
Beta	0.95	0.98	−0.03	(−1.40)
Idiosyncratic volatility	33.7%	33.26%	0.44%	(0.86)
CAR (annual EA)	1.76%	−0.33%	2.09%	(18.94)
Dividend payment	63.8%	64.28%	−0.53%	(−2.96)
Ret (m-3, m-1)	15.0%	12.1%	3.0%	(2.12)
ROA	6.3%	2.9%	3.3%	(13.39)
Leverage	43.8%	48.7%	−4.9%	(−13.00)
BM	0.38	0.89	−0.52	(−7.43)
EP	4.2%	5.1%	−0.9%	(−1.98)
Accrual	1.9%	−1.1%	3.0%	(11.90)
SOE	1.9%	−1.1%	3.0%	(11.90)
N (annual average)	219	1725		
	Suspicious	Nonsuspicious	Difference	t-stat
Panel B. Comparison of suspicious splits with nonsuspicious splits				
Size (in millions RMB)	$5491	$6777	-$1285	(−2.77)
Price	$18.00	$17.41	0.59	(1.19)
Analyst	10.5	10.3	0.3	(0.47)
Turnover (qtr)	37.6%	39.7%	−2.2%	(−1.11)
Beta	0.92	0.96	−0.04	(−3.43)
Idiosyncratic volatility	32.85%	33.81%	−0.96%	(−1.50)
CAR (annual EA)	2.06%	1.68%	0.38%	(1.58)
Dividend payment	62.96%	64.01%	−1.05%	(−0.33)
Ret (m-3, m-1)	5.7%	16.7%	−11.1%	(−4.34)
ROA	6.6%	6.2%	0.5%	(1.35)
Leverage	41.9%	44.7%	−2.7%	(−1.75)
BM	0.34	0.39	−0.05	(−5.16)
EP	3.8%	4.3%	−0.5%	(−2.69)
Accruals	5.3%	1.2%	4.1%	(5.15)
SOE	0.0%	69.5%	−69.5%	(−19.03)
N (annual average)	46	172		

- This supports the paper's claim that warning signs may not be easy for unsophisticated retail investors to process.

5.3 Table 2: unconditional abnormal returns around split announcements

Read:

- All splits have a positive announcement-month abnormal return of about 4.45% and a positive three-day announcement return of about 1.85%.
- Non-SOE splits show a larger announcement reaction than SOE splits.
- The key contrast is the longer-run reversal: non-SOE splits have a negative abnormal return of about -5.31% from month +4 to +18, while SOE splits have almost no reversal.
- This motivates the focus on non-SOE suspicious splits.

5.4 Table 3: suspicious versus regular splits

Read:

- Suspicious splits have an announcement-month abnormal return of 5.69% and a further 3.26% drift in months +1 to +3.
- They then reverse by -10.99% from months +4 to +18.
- Regular non-SOE splits have a positive announcement reaction and drift but only a very small and insignificant reversal.

Table 2
Abnormal returns around split announcement: SOE vs. non-SOE splits.
Table 2 reports the average buy-and-hold abnormal returns around split announcements for all split announcements and SOE/non-SOE split announcers. The average buy-and-hold abnormal return is calculated as the buy-and-hold return minus the size-decile benchmark where $t = 0$ is the calendar month (day) of the split announcement. Panel A (B) reports the monthly (daily) abnormal returns around all stock split announcements. t-statistics (presented in parentheses) are calculated using robust (White) standard errors that are clustered by calendar month. The sample period is from 1999/01 to 2015/06.

	N		[−3 to −1]	[month 0]	[+1 to +3]	[+4 to +18]	[0 to +18]
Panel A. Monthly abnormal returns around split announcements							
All splits	3716	mean	4.75	4.45	2.15	−2.47	4.77
		t-stat	(6.25)	(13.98)	(3.31)	(−1.59)	(3.11)
Non-SOE	1712	mean	3.91	4.87	2.99	−5.31	3.55
		t-stat	(3.71)	(9.74)	(3.53)	(−2.17)	(1.47)
SOE	2004	mean	5.47	4.09	1.43	−0.05	5.81
		t-stat	(7.35)	(11.30)	(1.84)	(−0.03)	(3.45)
	N		[−10, −2]		[−1,+1]		[+2,+10]
Panel B. Daily abnormal returns around split announcements							
All splits	3716	mean	2.66		1.85		−0.03
		t-stat	(11.47)		(13.62)		(−0.20)
Non-SOE	1712	mean	2.88		2.04		0.22
		t-stat	(8.34)		(10.17)		(0.87)
SOE	2004	mean	2.47		1.69		−0.25
		t-stat	(11.80)		(10.68)		(−1.04)

Table 3
Suspicious splits: Monthly abnormal returns around split announcement.
Table 3 reports the average monthly buy-and-hold abnormal returns around split announcements for suspicious splits. The average buy-and-hold abnormal return is calculated as the buy-and-hold return minus the size-decile benchmark where month=0 is the calendar month of the split announcement. Panel A presents suspicious splits and regular non-SOE splits. Panel B presents subsamples of suspicious split announcers by type. Lock-up expiration is non-SOE split announcers that have lock-up expirations that occur in months −1 to +6 around the split announcement of private placements or restricted shares held by influential shareholders. The sample period for the lock-up analysis is from 2006/01 to 2015/06 because reporting on lock-ups begins in 2006. Atypical timing is non-SOE split announcers that were either (1) in the bottom quintile in returns during the previous three months or (2) announced a split outside of an earnings announcement period. High accrual is non-SOE split announcers that are in the top quintile of accruals. t-statistics (presented in parentheses) are calculated using robust (White) standard errors that are clustered by calendar month. The sample period is from 1999/01 to 2015/06.

	N		[−3 to −1]	[month 0]	[+1 to +3]	[+4 to +18]	[0 to +18]
Panel A. Suspicious vs. regular splits (Non-SOE firm sample)							
Suspicious	787	mean	−2.25	5.69	3.26	−10.99	−0.58
		t-stat	(−1.92)	(6.26)	(2.39)	(−3.40)	(−0.18)
Regular non-SOE	925	mean	9.14	4.17	2.76	−0.47	7.07
		t-stat	(8.05)	(7.77)	(3.77)	(−0.14)	(2.13)
	N		[−3 to −1]	[month 0]	[+1 to +3]	[+4 to +18]	[0 to +18]
Panel B. Suspicious splits by type							
Lock-up expiration	146	mean	0.50	4.71	4.72	−13.63	−2.18
		t-stat	(0.32)	(3.69)	(2.09)	(−2.98)	(−0.49)
Atypical timing	468	mean	−7.85	6.10	3.49	−9.74	1.33
		t-stat	(−4.20)	(5.04)	(1.86)	(−2.51)	(0.33)
High accruals	344	mean	2.56	5.35	2.92	−14.64	−5.09
		t-stat	(1.85)	(4.57)	(2.71)	(−3.54)	(−1.26)

- The three suspicious categories all show the same broad pattern: short-run appreciation followed by long-run decline.

5.5 Figure 2: retail attention and suspicious split returns

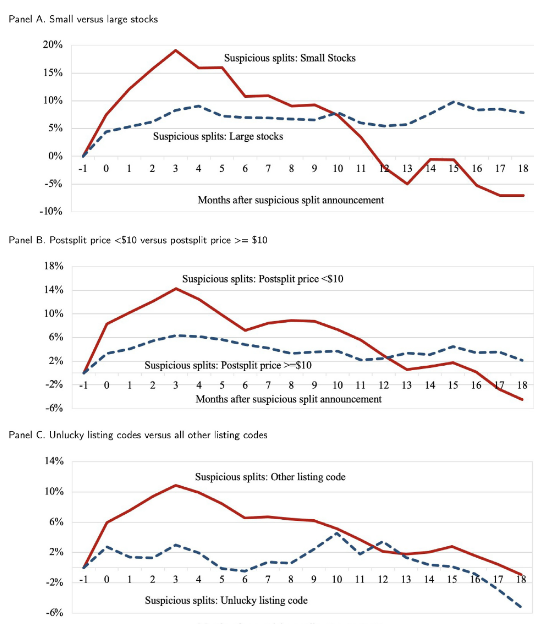


Fig.. 2. The effect of retail attention on returns of suspicious splits.
Fig. 2 plots the average monthly buy-and-hold abnormal returns (BHAR) of subsamples of suspicious splits with high and low retail attention. Panel A plots the BHAR of suspicious splits of small (large) stocks formed using the bottom 30 (top 70) percentile of market capitalization ranked on the previous quarter breakpoints. Panel B plots the BHAR of suspicious splits that experience a drop in the postsplit nominal price to less than $10 and those for which the postsplit nominal price remains >=$10. Panel C plots the BHAR of suspicious splits with unlucky listing codes and other listing codes that do not satisfy the unlucky criteria. An unlucky listing code contains the unlucky digit 4 but not any of the lucky digits 6, 8, or 9. Suspicious splits are split announcements by firms that meet one or more of the following criteria: lock-up expirations, atypical timing, or high accruals. The sample period is from 1999/01 to 2015/06.
Fig. 2. Continued
Panel A. Small versus large stocks
Panel B. Postsplit price < $10 versus postsplit price >= $10
Panel C. Unlucky listing codes versus all other listing codes.

Table 4
Trader net buying activity around suspicious split announcement.
Table 4 reports the average cumulative net buying in the preannouncement (t-20 to −1) and postannouncement (t = 0 to +60) period for each investor type for split announcements on the Shanghai Stock Exchange during the period from 1999/01 to 2015/06. Small retail is trading accounts with <$5 million RMB. Large accounts are trading accounts >=$5 million RMB. Institutional investors include mutual funds, hedge funds, and other types of institutional investors. Other investors include qualified foreign investors and social security accounts. Nonsuspicious splits consist of both regular firms and SOE firms that announced splits, but not classified as suspicious. t-statistics (presented in parentheses) are calculated using robust (White) standard errors that are clustered by calendar month.

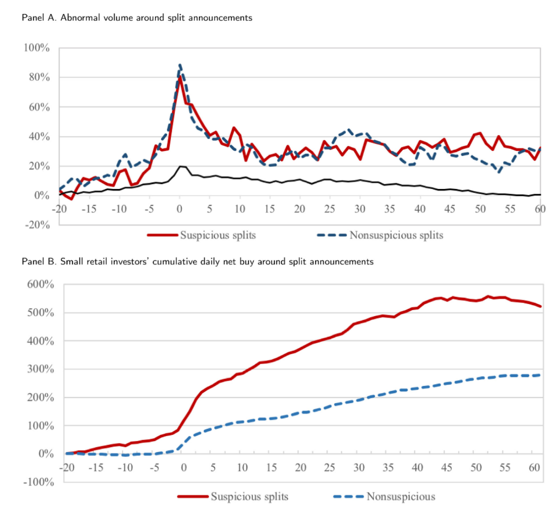
	Small retail		Large accounts		Institutional investors		Other investors	
	[−20 to −1]	[0 to +60]	[−20 to −1]	[0 to +60]	[−20 to −1]	[0 to +60]	[−20 to −1]	[0 to +60]
[A] Suspicious split	0.84	4.45	0.19	−0.85	−1.06	−3.43	0.03	−0.17
	(3.09)	(3.72)	(1.95)	(−2.74)	(−3.34)	(−3.62)	(0.69)	(−1.47)
[B] Nonsuspicious split	0.17	2.64	0.18	−0.25	−0.36	−2.35	0.01	−0.04
	(1.79)	(4.72)	(7.39)	(−4.87)	(−3.73)	(−4.82)	(0.50)	(−0.62)
Subsample of nonsuspicious								
[C] Non-SOE split	0.20	4.24	0.24	−0.44	−0.48	−3.53	0.04	−0.27
	(1.40)	(4.17)	(5.38)	(−3.29)	(−2.95)	(−3.82)	(0.86)	(−2.01)
[D] SOE split	0.16	2.19	0.16	−0.20	−0.33	−2.01	0.01	0.02
	(1.34)	(4.25)	(6.06)	(−3.92)	(−2.81)	(−4.44)	(0.24)	(0.26)
Difference between suspicious and nonsuspicious splits								
[A]-[B]	0.67	1.81	0.01	−0.59	−0.69	−1.09	0.01	−0.13
Suspicious-nonsuspicious	(2.34)	(1.85)	(0.07)	(−1.98)	(−2.04)	(−1.41)	(0.33)	(−1.23)
[A]-[C]	0.64	0.21	−0.05	−0.41	−0.57	0.10	−0.01	0.10
Suspicious-non-SOE	(2.20)	(0.20)	(−0.47)	(−1.30)	(−1.59)	(0.11)	(−0.25)	(0.69)
[A]-[D]	0.68	2.26	0.02	−0.65	−0.73	−1.42	0.02	−0.19
Suspicious-SOE	(2.26)	(2.15)	(0.24)	(−2.12)	(−2.09)	(−1.71)	(0.55)	(−1.69)

Read:

- Figure 2 shows that the suspicious-split return pattern is stronger among small stocks and stocks whose postsplit price falls below RMB 10.
- These are exactly the stocks likely to draw stronger retail attention.

- The pattern is weaker for unlucky listing codes, which some retail investors avoid because of numerological superstition.
- This heterogeneity supports the attention-based mechanism rather than a purely mechanical split effect.

5.6 Figure 3 and Table 4: who buys and who sells?



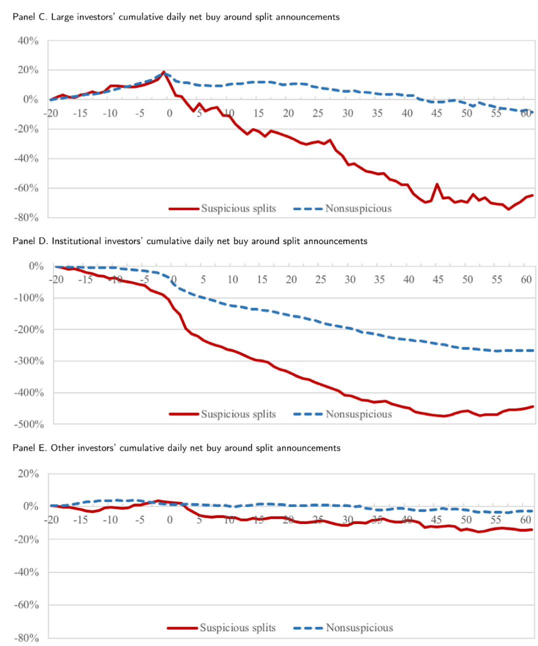


Fig. 3. Abnormal volume and cumulative daily net buying by investor groups around split announcement.
Fig. 3 plots abnormal volume and the average cumulative daily net buying by investor type around the announcement of a stock split for stocks on the Shanghai Stock Exchange. Panel A plots abnormal volume around the announcement of suspicious splits, nonsuspicious splits, and earnings with no concurrent split. Abnormal volume is defined as the daily dollar volume divided by the average daily dollar volume over the past year. Panels B, C, D, and E plot the cumulative net buying of small retail investors with trading accounts <$5 million RMB, large accounts with accounts >=$5 million RMB, institutional investors, and other investors, respectively. We report average cumulative net buying for suspicious splits (solid red line) and nonsuspicious splits, which consist of both regular + SOE splits (dashed blue line). Net buy is the total buy minus sell volume for each investor group divided by average daily volume over the past year. The sample consists of all stock transfers on the SSE announced between 1999/01 and 2015/06.
Panel A. Abnormal volume around split announcements
Panel B. Small retail investors' cumulative daily net buy around split announcements
Fig. 3 Continued
Panel C. Large investors' cumulative daily net buy around split announcements
Panel D. Institutional investors' cumulative daily net buy around split announcements
Panel E. Other investors' cumulative daily net buy around split announcements.

Fig. 3. Continued

Read:

- Figure 3 shows that abnormal volume rises before and after split announcements for both suspicious and nonsuspicious splits.
- The key difference is in trading direction: small retail investors accumulate suspicious splits, while large accounts and institutions sell.
- Table 4 quantifies this. Small retail investors buy suspicious splits both before and after the announcement, with postannouncement cumulative net buying around 4.45 units of average daily volume.
- Large accounts and institutional investors are net sellers after suspicious split announcements.
- The evidence is consistent with a wealth transfer from less sophisticated retail investors to more sophisticated investors and insiders.

5.7 Table 5: account characteristics of suspicious-split buyers

Read:

- Table 5 asks whether less sophisticated individual investors are more likely to buy suspicious splits.

- Investors with smaller accounts, weaker performance, and less experience are more likely to purchase suspicious splits.
- Younger investors and male investors are also more likely to buy them.
- The results survive restricting the sample to accounts that buy at least one split, so the evidence is not merely about who buys splits in general.

Table 5
Are less sophisticated investors more likely to buy suspicious splits?
Table 5 reports results from a Poisson regression of the number of suspicious splits purchased on retail investor characteristics from the Shanghai Stock Exchange using a random sample of individual accounts during the period from 2013/01 to 2015/06. Column 1 analyzes all the accounts in the random sample. Column 2 analyzes a subsample of accounts that purchased a stock split. Wealth is the natural logarithm of the average monthly account value in RMB. Return performance is the average monthly return performance calculated by accumulating the daily return of positions held from the prior day excluding all suspicious split holdings. Experience is the number of months the account has been open until the beginning of our sample period (Jan 2013). If the starting month is after Jan 2013, we set the value equal to zero. # of stocks bought is the average monthly number of stocks bought. # of stocks held is the average monthly number of stocks held in the account. Age is the age of the account holder at Jan 2013. Female is a dummy variable equal to one if the account holder is female, and zero otherwise. t-statistics (presented in parentheses) are calculated using robust (White) standard errors.

Sample:	All retail accounts with transactions	Subsample of retail accounts with split purchases
Investor sophistication		
Wealth	−1.871	−1.650
	(−9.31)	(−6.44)
Return performance	−0.137	−0.011
	(−30.95)	(−3.09)
Experience	−0.049	−0.037
	(−14.99)	(−11.02)
Transaction activity		
# of stocks bought	0.042	0.028
	(75.88)	(36.75)
# of stocks held	0.005	0.003
	(3.65)	(1.72)
Demographics		
Age	−0.003	−0.003
	(−3.26)	(−3.37)
Female	−0.124	−0.059
	(−7.42)	(−3.55)
Intercept	−3.655	−1.146
	(−73.73)	(−22.85)
Pseudo R-squared	0.057	0.023
Observations	123,160	35,716

Table 6
Who are the beneficiaries?
Table 6 reports results from a monthly panel regression of block trades (columns 1 and 2) and share pledge loans (columns 3 and 4) initiated as a percentage of market capitalization on the recent announcement of a suspicious split. Block trades is defined as the total monthly value of shares traded through block trade as a percentage of market capitalization. Block trades have transaction amounts greater than $2 million RMB in value or 300,000 shares. Share pledge loans is defined as the total monthly value of shares initially pledged as loan collateral as a percentage of market capitalization. Suspicious split is a dummy variable equal to one if the suspicious split is announced in the current or preceding month, and zero otherwise. Split is a dummy variable equal to one if the split is announced in the current or preceding month, and zero otherwise. Lock-up expiration is a dummy variable equal to one if the split is announced in the current or preceding month and has a lock-up expiration in the month −1 to +6 around the split announcement. Atypical timing is a dummy variable equal to one if the split is announced in the current or preceding month and the stock was in the bottom quintile of returns during the previous three months or the split is announced outside of an earnings announcement, and zero otherwise. High accrual is a dummy variable equal to one if the split was announced in the current or preceding month and was in the top quintile of accruals, where accruals is defined as net profit minus operation expense. The regressions include year-month fixed effects and industry-year fixed effects. t-statistics (presented in parentheses) are calculated using robust standard errors that are clustered by industry-year. Appendix A provides variable definitions. The sample period for block trades is from 2002/01 to 2015/07. The sample period for pledge loans is from 2006/01 to 2015/07.

	(1) Block trades	(2) Block trades	(3) Pledge loans	(4) Pledge loans
Suspicious split	0.89		0.02	
	(2.81)		(2.02)	
Suspicious split: Lock-up expiration		1.19		−0.01
		(2.12)		(−0.61)
Suspicious split: Atypical timing		1.14		0.01
		(2.05)		(0.80)
Suspicious split: High accrual		−0.02		0.03
		(−0.03)		(2.25)
Split	0.09	0.09	0.00	0.00
	(0.92)	(0.91)	(0.61)	(0.83)
Lock-up expiration	2.49	2.49	0.00	0.00
	(4.53)	(4.53)	(1.34)	(1.45)
High accrual	−0.04	−0.04	0.00	0.00
	(−1.22)	(−1.18)	(0.75)	(0.63)
Ret (m-3, m-1)	0.43	0.44	0.00	0.00
	(3.83)	(3.84)	(1.63)	(1.60)
SOE	−0.52	−0.52	−0.03	−0.03
	(−6.65)	(−6.56)	(−4.45)	(−4.45)
log(Size)	−0.13	−0.13	0.01	0.01
	(−5.68)	(−5.69)	(3.89)	(3.89)
ROA	0.01	0.01	−0.00	−0.00
	(0.84)	(0.84)	(−7.27)	(−7.27)
BM	−0.17	−0.17	−0.01	−0.01
	(−3.19)	(−3.19)	(−3.72)	(−3.73)
CAR (3-day)	0.03	0.04	0.01	0.01
	(0.18)	(0.20)	(1.88)	(1.86)
Turnover	−0.05	−0.05	−0.00	−0.00
	(−0.67)	(−0.67)	(−1.78)	(−1.78)
Investment	0.02	0.02	−0.01	−0.01
	(0.25)	(0.23)	(−1.82)	(−1.82)
Leverage	−0.27	−0.27	0.01	0.01
	(−2.33)	(−2.32)	(2.25)	(2.24)
Age	−0.45	−0.45	0.00	0.00
	(−6.06)	(−6.07)	(1.53)	(1.54)
Dividend amount	−1.79	−1.80	−0.26	−0.26
	(−1.64)	(−1.65)	(−3.78)	(−3.76)
Constant	5.12	5.12	−0.08	−0.08
	(8.57)	(8.58)	(−3.04)	(−3.04)
R-squared	0.0330	0.0330	0.023	0.023
Observations	266,269	266,269	209,638	209,638
Sample period	2002/01–2015/07	2002/01–2015/07	2006/01–2015/07	2006/01–2015/07

5.8 Table 6: who benefits from suspicious splits?

Read:

- Table 6 links suspicious split announcements to two potential insider-benefit channels: block trades and share pledge loans.
- Suspicious splits are associated with higher block trades shortly after announcement.
- The block-trade relation is especially strong for suspicious splits with lock-up expirations and atypical timing.
- High-accrual suspicious splits are associated with share pledge loans, consistent with insiders using temporarily inflated stock prices as collateral.
- Ordinary split announcements do not show the same relation, which supports the suspicious-split interpretation.

5.9 Table 7: message-board activity before suspicious splits

Read:

- Table 7 estimates Poisson regressions of Eastmoney Guba message-board activity.
- Message-board activity rises significantly in the 14 days before suspicious split announcements.
- The estimated incidence-rate ratios imply that online activity is around 1.7 to 2.1 times higher before suspicious splits, depending on the outcome and controls.
- This evidence is circumstantial but consistent with a broader pump-and-dump effort that uses public attention channels.

Table 7
Stock message board activity preceding suspicious split announcements.
Table 7 reports estimates from a Poisson regression of message board activity on the period preceding the announcement of suspicious splits. The posts are from the Eastmoney Guba message board and contains 789,461 total postings and 1410 stocks. The dependent variables are # of title characters, # of post characters, and # of posts. Suspicious preperiod is a dummy that equal to one if a suspicious split announcement will occur in the next 14 days, and zero otherwise. Nonsuspicious preperiod is a dummy that equal to one if a nonsuspicious split will occur in the next 14 days, and zero otherwise. Earnings preperiod is a dummy that equal to one if an earnings announcement will occur in the next 14 days, and zero otherwise. The regression includes firm fixed effects. t-statistics (presented in parentheses) are calculated using robust standard errors that are clustered by date. Appendix A provides variable definitions. The sample period starts in 2010/01 and ends in 2013/03.

	(1) # of title characters	(2) # of title characters	(3) # of title characters	(4) # of post characters	(5) # of posts
Suspicious preperiod	1.91	2.06	1.72	1.68	1.71
	(4.20)	(5.36)	(3.05)	(3.05)	(3.08)
Nonsuspicious preperiod			1.26	1.34	1.24
			(1.30)	(1.64)	(1.20)
Earnings preperiod			1.39	1.39	1.39
			(2.03)	(2.02)	(2.02)
Market cap (m-1)		0.12	0.12	0.10	0.12
		(−17.63)	(−17.26)	(−18.21)	(−17.00)
Return (m-1)		25.03	25.05	25.86	25.18
		(7.91)	(8.12)	(8.18)	(8.13)
Return (m-2)		52.36	48.89	53.97	49.36
		(13.76)	(13.18)	(12.43)	(13.39)
Return (m-3)		38.69	36.44	27.86	38.69
		(9.09)	(8.99)	(8.07)	(9.17)
Turnover (m-1)		0.71	0.71	0.72	0.70
		(−2.41)	(−2.53)	(−2.58)	(−2.58)
Turnover (m-2)		0.57	0.59	0.61	0.58
		(−3.50)	(−3.56)	(−3.44)	(−3.58)
Turnover (m-3)		0.94	0.93	0.99	0.91
		(−0.57)	(−0.73)	(−0.16)	(−0.85)
Observations	1156,626	1156,626	1156,626	1156,626	1156,626
Pseudo R-squared	0.0247	0.115	0.118	0.112	0.120

Table 8
Robustness test: Monthly abnormal returns around suspicious split announcements using alternative return benchmarks.
Table 8 reports the average monthly buy-and-hold abnormal returns around split announcements using a matched sample approach. Panel A matches each suspicious split to a sample of nonsplit stocks in the same quintile of market capitalization in month t−1 and with the same suspicious characteristic. Panel B matches each suspicious split to a sample of nonsplit stocks in the same quintile of market capitalization in month t−1 and same quintile of earnings surprise defined as the three-day market reaction to earnings announcement in the same month. Panel C matches each suspicious split to a sample of nonsplit stocks in the same quintiles of market capitalization in month t−1 and that announced a cash dividend in the same month. Month t = 0 is the calendar month of the split announcement. t-statistics (presented in parentheses) are calculated using robust (White) standard errors that are clustered by calendar month. The sample includes splits announced concurrently with earnings announcements during the period from 1999/01 to 2015/06.

	N		[−3 to −1]	[month 0]	[+1 to +3]	[+4 to +18]	[0 to +18]
Panel A. Alternative BHAR: Size/suspicious characteristic benchmark							
Suspicious*	655	mean	2.60	4.72	1.43	−13.13	−6.14
		t-stat	(3.89)	(5.46)	(1.33)	(−3.40)	(−1.59)
Suspicious types							
Lock-up expiration	146	mean	−1.10	4.62	5.51	−9.37	2.50
		t-stat	(−0.66)	(3.89)	(2.79)	(−1.90)	(0.62)
Atypical timing	303	mean	0.11	4.41	−0.02	−6.23	−1.71
		t-stat	(0.34)	(5.25)	(−0.02)	(−1.52)	(−0.34)
High accruals	322	mean	1.00	5.37	1.77	−23.76	−15.61
		t-stat	(0.72)	(4.01)	(1.25)	(−4.27)	(−2.87)
	N		[−3 to −1]	[month 0]	[+1 to +3]	[+4 to+18]	[0 to +18]
Panel B. Alternative BHAR: Size/earnings surprise benchmark							
Suspicious*	554	mean	−5.89	2.74	1.67	−14.10	−8.99
		t-stat	(−5.43)	(3.41)	(1.62)	(−2.95)	(−1.89)
Suspicious types							
Lock-up expiration	109	mean	0.01	3.52	5.24	−19.74	−8.87
		t-stat	(0.00)	(2.58)	(1.72)	(−3.82)	(−1.90)
Atypical timing	256	mean	−18.45	1.62	−0.28	−7.85	−6.56
		t-stat	(−16.97)	(1.50)	(−0.20)	(−1.37)	(−1.08)
High accruals	285	mean	1.00	3.40	2.51	−19.92	−13.12
		t-stat	(0.71)	(3.61)	(2.10)	(−3.84)	(−2.58)
	N		[−3 to −1]	[month 0]	[+1 to +3]	[+4 to +18]	[0 to +18]
Panel C. Alternative BHAR: Size/cash dividend benchmark							
Suspicious*	531	mean	−6.01	4.36	1.85	−9.09	−2.07
		t-stat	(−5.81)	(4.18)	(1.74)	(−2.50)	(−0.60)
Suspicious types							
Lock-up expiration	106	mean	−0.97	5.21	5.67	−15.44	−2.34
		t-stat	(−0.49)	(2.65)	(1.88)	(−2.76)	(−0.55)
Atypical timing	257	mean	−17.84	3.42	0.15	−2.00	1.89
		t-stat	(−16.13)	(2.84)	(0.11)	(−0.45)	(0.41)
High accruals	262	mean	0.72	4.85	2.25	−17.30	−9.34
		t-stat	(0.53)	(3.90)	(1.76)	(−3.71)	(−2.02)

5.10 Table 8: alternative matched benchmarks

Read:

- Table 8 shows that the main return pattern survives using matched nonsplit benchmarks.
- Matching on size and suspicious characteristics still gives a large reversal of about -13.13% from month +4 to +18.
- Matching on size and earnings surprise gives a reversal of about -14.10%.
- Matching on size and dividend announcements gives a reversal of about -9.09%.
- These results rule out the simple explanation that the suspicious-split pattern is caused only by earnings news, dividends, or the suspicious characteristics themselves.

6 Short summary

- **Method.** The paper combines event-study returns, investor-level trading records, account-characteristics regressions, insider-related transaction data, and message-board activity.
- **Identification.** The key design is an ex ante suspicious-split classification based on lock-up expirations, atypical timing, and high accruals, combined with comparisons to regular splits, SOE splits, and matched nonsplit benchmarks.
- **Main evidence.** Suspicious splits rise in price around the announcement and then reverse; small retail investors buy while sophisticated investors sell; less sophisticated accounts are more likely to buy suspicious splits.
- **Insider channel.** Suspicious splits are connected to block sales and share pledge loans, giving insiders concrete ways to benefit from temporary stock price inflation.
- **Main conclusion.** Stock splits can be used as part of a manipulation strategy in markets where retail investors dominate trading and short-sale constraints limit arbitrage.

7 Conclusion

7.1 Core conclusion

- The paper provides evidence that some Chinese firms used stock splits to manipulate retail investors.
- The evidence is not based on one pattern alone. It is based on a linked set of facts: suspicious pre-event characteristics, temporary price increases, long-run reversals, retail buying, sophisticated selling, insider-related transactions, and abnormal message-board activity.
- The paper shows that corporate actions can be misleading signals when investors are unsophisticated and arbitrage is limited.

7.2 Economic intuition

- A stock split is mechanically value-neutral, but it is highly salient.
- Salience attracts retail attention, especially in small, low-price stocks.
- If retail investors buy without understanding warning signs, the split can temporarily push prices above fundamentals.
- Informed investors, large shareholders, or insiders can exploit that temporary overvaluation by selling shares or pledging shares as collateral.
- Limited short selling weakens the arbitrage force that would otherwise push prices back down quickly.

7.3 Takeaway

This paper shows how a standard corporate action can become a manipulation device in a market with retail dominance, attention-based trading, and short-sale constraints. The main lesson is that investor protection cannot focus only on whether a disclosure is technically public. If the market is dominated by investors who cannot process warning signs, insiders can use salient but superficially benign announcements to create temporary demand and transfer losses to less sophisticated traders.